



Gilberto Recupito

Postdoctoral Researcher

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👤 Personal Information

Full Name

Gilberto Recupito

Place of Birth

Salerno, Italy

Date of Birth

January 25th, 1997

Current Address

Via Roma 2 - 84084, Fisciano, Salerno, Italy

👛 Current Position

Postdoctoral Researcher

Vrije Universiteit Brussel - With Prof. Coen De Roover

Feb 2026 – current

Brussels, Belgium

🕒 Work Experience

Research Intern

École Polytechnique de Montréal - With Prof. Foutse Khomh

Mar 2025 – May 2025

Montréal, Canada

Guest Researcher

German Aerospace center (DLR) - With Prof. Michael Felderer

May 2024 – Jul 2024

Cologne, Germany

Research Assistant

Tampere University (CloudSea.AI) - With Prof. Davide Taibi

Apr 2022 – Sep 2022

Tampere, Finland

🎓 Education

Ph.D. in Computer Science

Università degli studi di Salerno

Excellent, Advisors: Profs. Fabio Palomba & Dario Di Nucci

2025

Master's Degree (MSc) in Computer Science

Università degli studi di Salerno

Magna cum laude, Advisor: Prof. Fabio Palomba

2022

Research Areas

My primary research area is **Software Engineering for Artificial Intelligence (SE4AI)**. My work advances the engineering of AI-intensive systems by:

- › **AI Technical Debt:** Characterizing and mitigating sources of technical debt in AI-enabled systems, including system-level [13], data-centric [14], and ML-specific code quality issues [7], to improve reliability and maintainability.
- › **MLOps and Engineering of ML Pipelines:** Defining practices, features, and tools that support the development, automation, and operation of robust ML pipelines [15]. This includes empirical analysis of ML projects, enabling the characterization of *ML-Model Producers* and *ML-Model Consumers* [2].
- › **LLM4SE and Agentic AI:** Investigating how Large Language Models can support software engineering tasks, with a focus on the design of prompt patterns to manage complexity in LLM-generated code [3] and support code quality evaluation and secure code generation [1].
- › **Empirical Software Engineering & Mining Software Repositories:** Contributing to empirical methods and MSR research, including educational work culminating in a book chapter to support instructors in teaching MSR techniques [8].

Research Projects & Grants

FWO Project: CodeGuard

2026

Actively contributed to the Flanders Research Foundation project. Activities include research and knowledge sharing to support LLM and agents secure code generation. I actively contributed with research contribution, workshop organization jointly with companies and academies and hackatons for dissemination.

PRIN Project: FRINGE

2023-2025

Actively contributed to the proposal and the research activities of the PRIN project FRINGE awarded by NextGeneration EU and the MUR.

PRIN Project: QUALAI

2022-2025

Actively contributed to the proposal and the research activities of the PRIN project QUALAI awarded by the MUR.

Professional Activities

Professional Recognitions

Organizing Activities

4th Workshop on Software Quality Assurance for Artificial Intelligence (SQA4AI) - Organizer 2027 |
Richmond, VA, USA

3rd Workshop on Software Quality Assurance for Artificial Intelligence (SQA4AI) - Organizer 2026 |
Limassol, Cyprus

Committee Activities

International Conference on Program Comprehension (ICPC) 2027
ERA Track (Dublin, Ireland)

International Conference on the Foundations of Software Engineering (FSE) 2027
Industry Track (Shenzhen, China)

International Conference on Automated Software Engineering (ASE) 2026
Artifact Evaluation Track (Munich, Germany)

International Workshop on Software Security Testing (SECUTE) 2026
Program Committee Member (Munich, Germany)

International Workshop on Trustworthy and Responsible aUtonomous SysTems (TRUST) 2026
Program Committee Member (Munich, Germany)

International Conference on Software Maintenance and Evolution (ICSME) 2026
Tool Demo and Data Showcase Track, Industry Track, Replication and Negative Results Track (Benevento, Italy)

International Conference on AI Engineering (CAIN) 2026
Poster Track Program Committee (Rio De Janeiro, Brazil)

International Workshop on Empirical Prompt Engineering for Software Engineering (Prompt-SE) 2026
Program Committee Member (Glasgow, UK)

1st ECAI Workshop on MLOps (MLOps) 2025
Session Chair (Bologna, Italy)

International Conference on AI Foundation Models and Software Engineering (FORGE) 2026, 2025, 2024
Program Committee (Rio De Janeiro, Brazil; Ottawa, Canada; Lisbon, Portugal)

Euromicro Conference on Software Engineering and Advanced Applications (SEAA) 2025
Session Chair (Salerno, Italy)

International Conference on Software Analysis, Evolution and Reengineering (SANER) 2025
Session Chair (Montréal, Canada)

2nd Workshop on Software Quality Assurance for Artificial Intelligence (SQA4AI) 2025
Program Committee Member (Montréal, Canada)

International Conference on Automated Software Engineering (ASE) 2024
Artifact Evaluation Track Program Committee (California, United States)

International Working Conference on Mining Software Repositories (MSR) <i>Junior Program Committee (Lisbon, Portugal)</i>	2024
Seminar Series on Advanced Techniques & Tools for Software Evolution (SATToSE) <i>Social Media Chair (Salerno, Italy)</i>	2023
International Conference on Software Engineering Advances (ICSEA) <i>Program Committee Member (Valencia, Spain)</i>	2023
Annual Symposium for Computer Science <i>Organizing Committee Member (Tampere, Finland)</i>	2022
Guest Editor	
Journal of Systems and Software (<i>Special Issue SQA for AI</i>)	2026
Journal Reviewer	
IEEE Transactions on Software Engineering	2025-2026
Journal of Systems and Software	2023-2026
Automated Software Engineering Journal	2025
SoftwareX	2024-2026
Journal of Software: Evolution and Process	2024
ACM Transactions on Software Engineering and Methodology	2023-2026
Empirical Software Engineering	2024-2026
IET Software	2025
ACM Conf. On Computer-Supported Cooperative Work And Social Computing	2023, 2024
Science of Computer Programming	2023
Journal of Decision Systems	2022

Teaching Activities

During these years, I served as Teaching Assistant for the following Bachelor's and Master's Courses:

Software Dependability (M.Sc.) <i>Prof. Di Nucci</i> Support to course projects.	2023, 2024
Software Engineering for Artificial Intelligence (M.Sc.) <i>Prof. Palomba</i> Support to course projects and lectures on Transfer Learning, Transformers and MLOps.	2023, 2024
Software Engineering, Management and Evolution (M.Sc.) <i>Prof. De Lucia</i> Support to course projects and lectures on DevOps.	2023, 2024
Fundamentals of Artificial Intelligence (B.Sc.) <i>Prof. Fabio Palomba</i> Student Tutor for course projects and lecture on MLOps and Software Quality for AI.	2024
Software Engineering (B.Sc.) <i>Prof. De Lucia</i>	2023, 2025

Support to course projects and lecturer for Pull-based development, Testing toolkit, and LLM-assisted development lectures.

Advising Activities

Currently, I am supporting Prof. Coen De Roover in the supervision of a Ph.D. Student on the thesis: *Analysis and Characterization of Policy-as-Code Adoption in Software Systems*. Additionally, in my career, I had the opportunity to advise Bachelors and Masters students in their career.

- A Comparative Analysis of Code Quality in Data Science Notebooks. *Melissa Njenga, Master's Degree (2026)*.
- CARLAConfigHub: A Tool for Automating Scenario Generation for the CARLA Autonomous Driving Simulator *Armando Vigliotti, Bachelor's Degree (2026)*.
- A Comparative Analysis of Code Review Activities in Human vs. AI-Generated Pull Requests on GitHub. *Samuele Nacchia, Bachelor's Degree (2026)*.
- Dynamic Prompt Injection in Large Language Models for Code Generation. *Francesco Niemiec, Bachelor's Degree (2026)*.
- Computational Sustainability: Measuring the Energy Impact of Language Models on Consumer Hardware. *Yevgeniy Terentyev, Master's Degree (2026)*.
- CRANE - Code Review AI Network Engine. *Luca Morelli, Master's Degree (2025)*.
- Development and testing of CodeSmile: a plugin for the automatic analysis of Machine Learning Code Smells. *Daniele Pio Scaparra, Bachelor's Degree (2025)*.
- Analysis of the Correlation between Data Smells, Performance, Fairness, and Sustainability in Machine Learning-Based Systems. *Domenico D'Antuono, Master's Degree (2025)*.
- Analysis of the evolution of code smells in Machine Learning-Based Systems. *Simone Silvestri, Master's Degree (2025)*.
- CAMILLE: A chatbot for ML-specific code Smells identification and refactoring. *Francesco Pinto, Bachelor's Degree (2024)*.
- Analysis and Exploration of MLOps tools for monitoring data drift - *Ranjeet Kumar, Bachelor's Degree (2024)*.
- Synthetic data smell injection: Effect and impact on quality in ML-enabled systems. *Davide La Gamba, Master's Degree (2024)*.
- Machine Learning Detection of Data Smells. *Nicolò Gallotta, Bachelor's Degree (2024)*.
- Multi-Agent System for information needs extraction of code review tasks. *Giulio Palladino, Bachelor's Degree (2024)*.
- MLOps: A systematic review of the literature on MLOps practices and properties. *Damiana Buono, Bachelor's Degree (2023)*.
- Data Smells: towards the exploration and analysis of the impact on data quality. *Raimondo Rapaciuolo, Master's Degree (2023)*.

Guest Lectures

Guest Lecture on LLMs for Software Engineering

Vrije Universiteit Brussel

Host: Prof. Coen De Roover

2026

Brussel, Belgium

Guest Lecture on Technical Debt

University of Salerno (M.Sc. Sustainable Software Engineering)

Host: Prof. Di Nucci

2025

Salerno, Italy

Guest Lectures on LLMs for Automated Software Development

University of Salerno (B.Sc. Software Engineering)

Host: Prof. De Lucia.

2023, 2024

Salerno, Italy

Guest Lectures on Maven, GitHub, and testing frameworks

University of Salerno (B.Sc. Software Engineering)

Host: Prof. De Lucia.

2023, 2024

Salerno, Italy

Publications

- [1] Alfonso Cannavale, Gilberto Recupito, Coen De Roover, Fabio Palomba, and Andrea De Lucia. "Smarter, but Not Safer: An Empirical Analysis of the Functional-Security Gap in Evolving LLMs". In: *International Conference on Software Maintenance and Evolution (ICSME)*. 2026.
- [2] Vincenzo De Martino, Gilberto Recupito, Giammaria Giordano, Filomena Ferrucci, Dario Di Nucci, and Fabio Palomba. "Into the ML-universe: An improved classification and characterization of machine-learning projects". In: *Journal of Systems and Software* (2025).
- [3] Antonio Della Porta, Gilberto Recupito, Stefano Lambiase, Dario Di Nucci, and Fabio Palomba. "Unlocking Code Simplicity: The Role of Prompt Patterns in Managing LLM Code Complexity". In: *International Workshop of Software Quality Assurance for Artificial Intelligence (SQA4AI)*. 2025.
- [4] Fabio Palomba, Gianmario Voria, Alessandra Parziale, Viviana Pentangelo, Antonio Della Porta, Vincenzo De Martino, Gilberto Recupito, and Giammaria Giordano. "Teaching Software Engineering for Artificial Intelligence: An Experience Report". In: *Euromicro Conference on Software Engineering and Advanced Applications*. Springer. 2025, pp. 214–230.
- [5] Gilberto Recupito, Vincenzo De Martino, Dario Di Nucci, and Fabio Palomba. "A First Look at the Lifecycle of DL-Specific Self-Admitted Technical Debt". In: *International Workshop of Software Quality Assurance for Artificial Intelligence (SQA4AI)*. 2025.
- [6] Gilberto Recupito, Giammaria Giordano, Dario Di Nucci, and Fabio Palomba. "Detecting Semantic Data Smells with BERT: A Transformer-Based Approach to Data Quality". In: (2025).
- [7] Gilberto Recupito, Giammaria Giordano, Filomena Ferrucci, Dario Di Nucci, and Fabio Palomba. "When code smells meet ML: on the lifecycle of ML-specific code smells in ML-enabled systems". In: *Empirical Software Engineering (EMSE)*. 2025.
- [8] Zadia Codabux, Fatemeh Fard, Roberto Verdecchia, Fabio Palomba, Dario Di Nucci, and Gilberto Recupito. "Teaching Mining Software Repositories". In: *Handbook on Teaching Empirical Software Engineering*. Springer, 2024.
- [9] Antonio Della Porta, Vincenzo De Martino, Gilberto Recupito, Carmine Iemmino, Gemma Catolino, Dario Di Nucci, and Fabio Palomba. "Using Large Language Models to Support Software Engineering Documentation in Waterfall Life Cycles: Are We There Yet?" In: (2024).

- [10] Davide La Gamba, Gerardo Iuliano, Gilberto Recupito, Giammaria Giordano, Filomena Ferrucci, Dario Di Nucci, and Fabio Palomba. "Toward a Search-Based Approach to Support the Design of Security Tests for Malicious Network Traffic". In: *Proceedings of the 28th International Conference on Evaluation and Assessment in Software Engineering*. EASE '24. , Salerno, Italy, Association for Computing Machinery, 2024. ISBN: 9798400717017. DOI: [10.1145/3661167.3661288](https://doi.org/10.1145/3661167.3661288).
- [11] Nicole Novielli, Rocco Oliveto, Fabio Palomba, Fabio Calefato, Giuseppe Colavito, Vincenzo De Martino, Antonio Della Porta, Giammaria Giordano, Emanuela Guglielmi, Filippo Lanubile, et al. "Continuous Quality Improvement of AI-based Systems: the QualAI Project". In: *18th ACM/IEEE International Symposium on Empirical Software Engineering and Measurement*. 2024.
- [12] Nicole Novielli, Rocco Oliveto, Fabio Palomba, Fabio Calefato, Giuseppe Colavito, Vincenzo De Martino, Antonio Della Porta, Giammaria Giordano, Emanuela Guglielmi, Filippo Lanubile, et al. "QualAI: Continuous Quality Improvement of AI-based Systems." In: *RCIS Workshops*. 2024.
- [13] Gilberto Recupito, Fabiano Pecorelli, Gemma Catolino, Valentina Lenarduzzi, Davide Taibi, Dario Di Nucci, and Fabio Palomba. "Technical debt in AI-enabled systems: On the prevalence, severity, impact, and management strategies for code and architecture". In: *Journal of Systems and Software* (2024), p. 112151. ISSN: 0164-1212. DOI: <https://doi.org/10.1016/j.jss.2024.112151>.
- [14] Gilberto Recupito, Raimondo Rapacciuolo, Dario Di Nucci, and Fabio Palomba. "Unmasking Data Secrets: An Empirical Investigation into Data Smells and Their Impact on Data Quality". In: *3rd International Conference on AI Engineering – Software Engineering for AI* (2024).
- [15] Gilberto Recupito, Fabiano Pecorelli, Gemma Catolino, Sergio Moreschini, Dario Di Nucci, Fabio Palomba, and Damian Andrew Tamburri. "A Multivocal Literature Review of MLOps Tools and Features". In: *Euromicro Conference on Software Engineering and Advanced Applications* (2022). DOI: <http://dx.doi.org/10.13140/RG.2.2.10257.71526>.

In Fede, August 27, 2026